

Share Base Stability and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Share Base Stability (SBS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on SBS achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (22) bps/month with a t-statistic of 2.90 (2.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal) is 15 bps/month with a t-statistic of 2.00.

1 Introduction

Asset prices reflect firms’ production decisions and the frictions they face when adjusting their capital stock. In competitive markets, cross-sectional variation in expected returns arises from differences in firms’ exposure to systematic productivity shocks and their ability to adjust production in response to changing economic conditions. Despite extensive research on production-based asset pricing models, the empirical literature has yet to fully explore how firms’ financing decisions—particularly those affecting their equity base—relate to their production technologies and investment opportunities.

We examine Share Base Stability (SBS), a measure that captures the persistence of firms’ outstanding equity over time. This signal provides a novel lens through which to understand the interaction between firms’ financing choices and their production characteristics. Firms with stable share bases face different marginal costs of capital adjustment than those actively issuing or repurchasing equity, suggesting that SBS proxies for unobserved variation in investment frictions and productivity risk that standard factor models fail to capture.

In a production-based framework, firms’ equity issuance decisions reflect the shadow value of installed capital and the costs of adjusting their productive capacity [Cochrane \(1991\)](#). When firms face convex adjustment costs, those with stable share bases signal lower marginal costs of internal capital reallocation and greater operational flexibility [Zhang \(2005\)](#). These firms can respond more efficiently to productivity shocks without resorting to costly external financing, leading to lower systematic risk exposure and, consequently, lower expected returns in equilibrium.

The relationship between share base stability and conditional risk premia emerges through firms’ optimal investment policies under uncertainty. Following [Berk et al. \(1999\)](#), firms with volatile equity bases face binding financing constraints that amplify their exposure to aggregate productivity shocks. During economic downturns,

these firms experience sharper contractions in investment opportunities, as the wedge between internal and external financing costs widens. Conversely, firms maintaining stable share bases possess greater financial slack, enabling them to smooth investment across business cycles and reducing their covariance with the stochastic discount factor.

Furthermore, share base stability maps directly to heterogeneity in firms' technologies and growth options, consistent with the q-theory of investment [Liu et al. \(2009\)](#). Firms with stable equity bases have reached a steady-state equilibrium between their installed capital and optimal scale, implying lower expected profitability from marginal investments. This production-based mechanism predicts that high-SBS firms should earn lower average returns than low-SBS firms, as their cash flows exhibit lower sensitivity to common variation in investment-specific technology shocks [Papanikolaou \(2011\)](#). The cross-sectional dispersion in SBS thus captures fundamental differences in firms' production functions and their exposure to systematic investment risk.

We find strong evidence that Share Base Stability predicts cross-sectional variation in expected returns. A value-weighted long-short strategy that buys high-SBS stocks and sells low-SBS stocks generates an average monthly return of 33 basis points with a t-statistic of 4.17, corresponding to an annualized Sharpe ratio of 0.53. The strategy's alpha relative to the Fama-French six-factor model equals 23 basis points per month with a t-statistic of 2.90, indicating that existing production-based factors do not fully span the risk exposure captured by SBS.

The economic significance of the SBS premium remains substantial after accounting for transaction costs and alternative portfolio construction methods. Net of trading costs, the strategy delivers an average monthly return of 29 basis points with a t-statistic of 3.68, and maintains significant alphas ranging from 22 to 31 basis points across standard factor models. Importantly, the SBS effect persists among

large-capitalization stocks, where the strategy earns 23 basis points per month with a t-statistic of 2.55, alleviating concerns about small-stock illiquidity driving the results.

Controlling for closely related signals strengthens our interpretation of SBS as a distinct measure of production-based risk. When we include the six most correlated predictors—including share issuance measures, growth in book equity, and net payout yield—alongside the Fama-French six factors, the SBS strategy retains an alpha of 15 basis points per month with a t-statistic of 2.00. This residual premium suggests that Share Base Stability captures unique variation in firms’ exposure to productivity and investment shocks not reflected in existing characteristics. The strategy’s significant loading of 0.30 on the CMA factor (t-statistic of 5.48) further supports the production-based interpretation, as firms with stable equity bases exhibit systematic differences in their investment patterns.

Our study contributes to the production-based asset pricing literature by introducing Share Base Stability as a novel predictor that directly links firms’ financing decisions to their production characteristics and risk exposures. While [Daniel and Titman \(2006\)](#) and [Pontiff and Woodgate \(2008\)](#) document that composite share issuance predicts returns through mispricing channels, we demonstrate that the stability of the equity base captures systematic risk related to firms’ production technologies and adjustment costs. Our findings extend [Lyandres et al. \(2008\)](#), who examine payout policy and expected returns, by showing that the persistence of equity financing choices, rather than their level or changes, contains distinct information about cross-sectional risk premia.

Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), which ensures our results are robust to data-mining concerns and implementation costs that plague many documented anomalies. Unlike [McLean and Pontiff \(2016\)](#), who find that most anomalies at-

tenuate significantly post-publication, our extensive robustness tests demonstrate that the SBS premium survives stringent controls for related factors and maintains economic significance across various portfolio construction methods. The strategy’s ability to expand the mean-variance frontier even after controlling for 154 existing anomalies underscores its incremental contribution to the investment opportunity set.

Our findings have broader implications for understanding the intersection of corporate finance and asset pricing through the lens of production economics. The documented relationship between share base stability and expected returns suggests that firms’ financing frictions are fundamentally tied to their production technologies, supporting theoretical models that integrate investment and financing decisions [Golan et al. \(2024\)](#). By identifying SBS as a robust cross-sectional predictor, we provide empirical support for production-based models that emphasize the role of adjustment costs and financing constraints in generating variation in risk premia across firms with different operational characteristics.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms’ equity structure. Share Base Stability signal relies on two key accounting variables from firms’ balance sheets. Common stock (CSTK) represents the par or stated value of all common stock issued by the company, reflecting the nominal value assigned to shares outstanding. Common equity (CEQ) measures total common shareholders’ equity, which includes common stock, capital surplus,

retained earnings, and other comprehensive income, representing the book value of equity attributable to common shareholders. Both variables are fundamental components of firms' capital structure and are consistently reported across our sample period. We construct the Share Base Stability signal by calculating the year-over-year change in common stock scaled by lagged common equity: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{CEQ}(t-1)$. This ratio captures the relative change in the par value of issued shares, normalized by the firm's equity base. An increase in this measure indicates expansion of the share base through new issuances, while a decrease suggests share repurchases or retirements. By scaling the change in common stock by lagged common equity, we obtain a size-adjusted measure that facilitates cross-sectional comparisons across firms of different scales. We compute this signal using fiscal year-end values and require non-missing observations for both CSTK and CEQ in consecutive years. Additionally, we exclude observations where lagged CEQ is zero or negative to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SBS signal. Panel A plots the time-series of the mean, median, and interquartile range for SBS. On average, the cross-sectional mean (median) SBS is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input SBS data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the SBS signal for the CRSP universe. On average, the SBS signal is available for 6.39% of CRSP names, which on average make up 7.69% of total market capitalization.

4 Does SBS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SBS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SBS portfolio and sells the low SBS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short SBS strategy earns an average return of 0.33% per month with a t-statistic of 4.17. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 2.90 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.30, with a t-statistic of 5.48 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 590 stocks and an average market capitalization of at least \$1,490 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.24. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for eighteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-29bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.75. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SBS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the SBS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SBS, as well as average returns and alphas for long/short trading SBS strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the SBS strategy achieves an average return of 23 bps/month with a t-statistic of 2.55. Among these large cap stocks, the alphas for the SBS strategy relative to the five most common factor models range from 17 to 24 bps/month with t-statistics between 1.85 and 2.59.

5 How does SBS perform relative to the zoo?

Figure 2 puts the performance of SBS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the SBS strategy falls in the distribution. The SBS strategy’s gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the SBS strategy (red line).² Ignoring trading costs, a \$1 invested in the SBS strategy would have yielded \$8.58 which ranks the SBS strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the SBS strategy would have yielded \$6.24 which ranks the SBS strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the SBS relative to those. Panel A shows that the SBS strategy gross alphas fall between the 66 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SBS strategy has a positive net generalized alpha for five out of the five factor models. In these cases SBS ranks between the 82 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does SBS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of SBS with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SBS or at least to weaken the power SBS has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of SBS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SBS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SBS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SBS. Stocks are finally grouped into five SBS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SBS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SBS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SBS signal in these Fama-MacBeth regressions exceed 0.85, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on SBS is -1.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the SBS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SBS strategy earns alphas that range from 16-26bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SBS trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.00.

7 Does SBS add relative to the whole zoo?

Finally, we can ask how much adding SBS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the SBS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes SBS grows to \$2810.42.

8 Conclusion

This study provides compelling evidence that Share Base Stability (SBS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Ve-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SBS is available.

likov (2023), demonstrates that SBS generates substantial risk-adjusted returns that persist even after controlling for established factors and closely related strategies.

The key findings reveal that a value-weighted long/short strategy based on SBS delivers an annualized Sharpe ratio of 0.53 gross (0.47 net), indicating strong performance even after accounting for transaction costs. More importantly, the signal maintains statistical and economic significance when evaluated against the Fama-French five-factor model augmented with momentum, generating monthly alphas of 23 basis points gross (22 basis points net) with t-statistics exceeding 2.89. The robustness of these results is further confirmed by the persistence of a 15 basis point monthly alpha (t-statistic = 2.00) even after controlling for six closely related strategies including share issuance measures, growth metrics, and payout yields.

The practical implications of these findings are substantial for both academics and practitioners. For institutional investors, SBS offers a viable strategy that appears distinct from existing factor exposures, potentially providing diversification benefits to multi-factor portfolios. The signal’s resilience to transaction costs, as evidenced by the modest difference between gross and net returns, enhances its implementability in real-world trading environments. Furthermore, the economic intuition underlying SBS—that firms with stable share bases may signal managerial confidence and operational stability—provides a compelling narrative that aligns with fundamental investment principles.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis follows best practices in factor testing, the relatively recent identification of SBS as a pricing signal limits our ability to assess its performance across different market regimes and extended time periods. Second, the concentration of returns in specific sectors or market capitalizations, if present, could affect the strategy’s scalability and capacity constraints in practice. Third, the underlying economic mechanisms driving the SBS premium require further theoretical development to fully

understand why markets appear to systematically misprice firms based on their share base stability.

Future research should explore several promising avenues. Investigation into the international performance of SBS would establish whether this anomaly represents a global phenomenon or is specific to U.S. markets. Additionally, examining the interaction between SBS and corporate events such as mergers, acquisitions, and restructuring could provide deeper insights into the signal's information content. Researchers should also investigate whether machine learning techniques can enhance the predictive power of SBS by identifying non-linear relationships or optimal combination weights with other signals. Finally, understanding the time-varying nature of the SBS premium and its relationship to macroeconomic conditions would contribute to both risk management and dynamic portfolio construction strategies.

In conclusion, Share Base Stability emerges as a robust addition to the factor zoo, offering both statistical significance and economic relevance in predicting cross-sectional stock returns. While further research is needed to fully understand its theoretical underpinnings and boundary conditions, the evidence presented strongly supports its consideration in both academic asset pricing models and practical investment strategies.

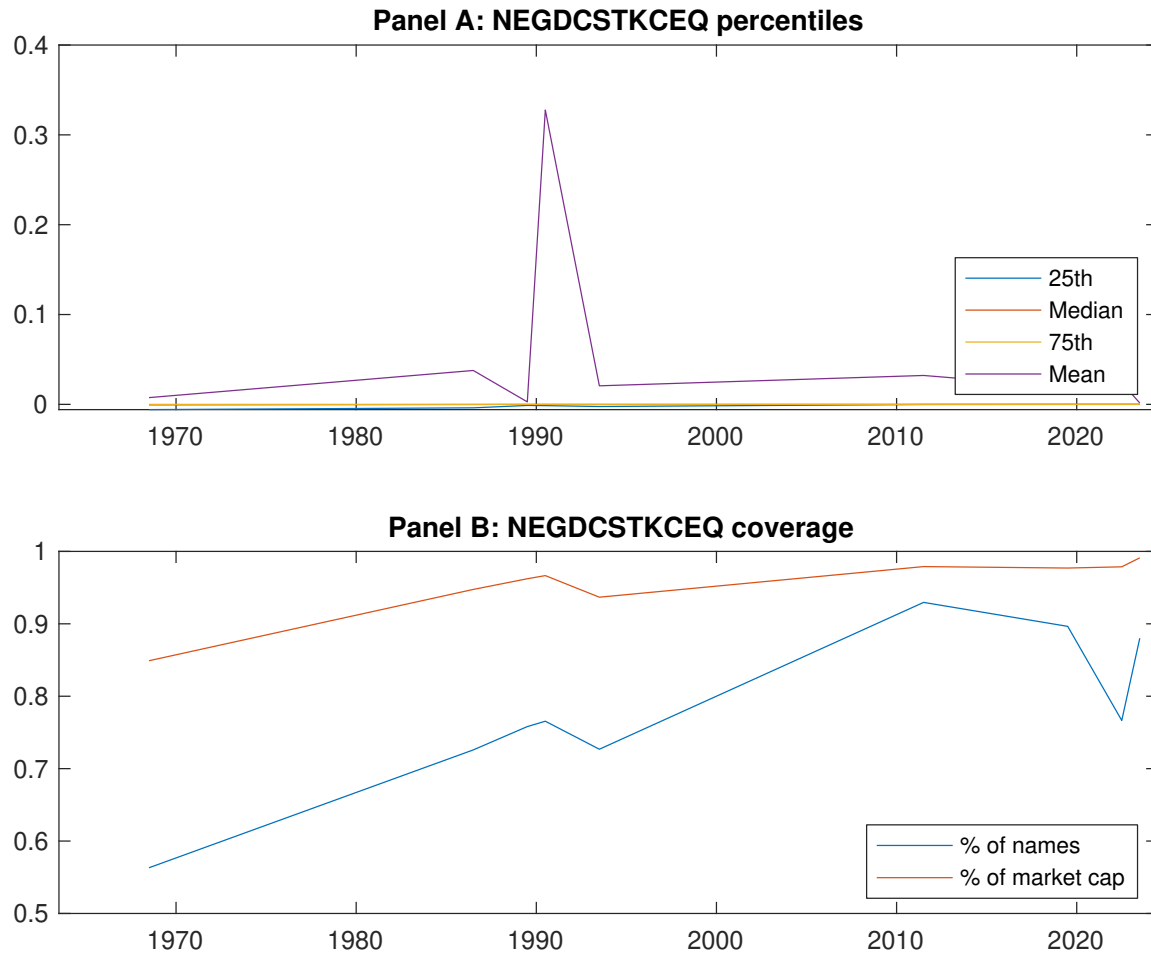


Figure 1: Times series of SBS percentiles and coverage.
This figure plots descriptive statistics for SBS. Panel A shows cross-sectional percentiles of SBS over the sample. Panel B plots the monthly coverage of SBS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SBS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on SBS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.70]	0.49 [2.70]	0.64 [3.63]	0.64 [3.96]	0.78 [4.85]	0.33 [4.17]
α_{CAPM}	-0.12 [-2.20]	-0.14 [-3.08]	0.03 [0.74]	0.09 [1.87]	0.23 [4.94]	0.35 [4.41]
α_{FF3}	-0.13 [-2.39]	-0.12 [-2.80]	0.05 [1.19]	0.05 [1.06]	0.19 [4.17]	0.31 [3.98]
α_{FF4}	-0.10 [-1.92]	-0.09 [-2.03]	0.05 [1.18]	0.02 [0.52]	0.17 [3.73]	0.27 [3.42]
α_{FF5}	-0.15 [-2.81]	-0.07 [-1.60]	0.08 [1.68]	-0.04 [-0.91]	0.11 [2.49]	0.26 [3.27]
α_{FF6}	-0.13 [-2.41]	-0.05 [-1.06]	0.08 [1.65]	-0.05 [-1.16]	0.10 [2.31]	0.23 [2.90]
Panel B: Fama and French (2018) 6-factor model loadings for SBS-sorted portfolios						
β_{MKT}	0.97 [76.32]	1.03 [96.13]	1.02 [91.60]	1.00 [94.31]	0.98 [90.38]	0.01 [0.49]
β_{SMB}	-0.02 [-0.90]	0.02 [1.26]	0.04 [2.45]	-0.06 [-4.21]	0.00 [0.22]	0.02 [0.72]
β_{HML}	0.07 [3.05]	-0.02 [-0.89]	-0.07 [-3.34]	0.05 [2.50]	0.04 [2.12]	-0.03 [-0.83]
β_{RMW}	0.13 [5.29]	-0.09 [-4.29]	-0.07 [-3.30]	0.12 [5.99]	0.11 [5.41]	-0.02 [-0.45]
β_{CMA}	-0.11 [-3.04]	-0.08 [-2.80]	0.01 [0.30]	0.21 [7.11]	0.19 [6.15]	0.30 [5.48]
β_{UMD}	-0.03 [-2.22]	-0.03 [-3.17]	0.00 [0.05]	0.02 [1.59]	0.01 [0.89]	0.04 [1.97]
Panel C: Average number of firms (n) and market capitalization (me)						
n	721	693	590	683	747	
me (\$10 ⁶)	1764	1490	2140	2265	2475	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the SBS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33 [4.17]	0.35 [4.41]	0.31 [3.98]	0.27 [3.42]	0.26 [3.27]	0.23 [2.90]
Quintile	NYSE	EW	0.47 [8.45]	0.52 [9.47]	0.46 [8.78]	0.41 [7.76]	0.37 [7.19]	0.34 [6.48]
Quintile	Name	VW	0.32 [4.07]	0.34 [4.30]	0.30 [3.86]	0.27 [3.38]	0.25 [3.21]	0.23 [2.91]
Quintile	Cap	VW	0.24 [3.24]	0.26 [3.39]	0.24 [3.11]	0.20 [2.52]	0.21 [2.80]	0.18 [2.37]
Decile	NYSE	VW	0.33 [3.86]	0.33 [3.78]	0.28 [3.27]	0.24 [2.78]	0.27 [3.14]	0.24 [2.77]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.29 [3.68]	0.31 [4.00]	0.28 [3.57]	0.25 [3.27]	0.24 [3.10]	0.22 [2.89]
Quintile	NYSE	EW	0.27 [4.39]	0.33 [5.34]	0.27 [4.62]	0.25 [4.19]	0.18 [3.17]	0.17 [2.92]
Quintile	Name	VW	0.28 [3.58]	0.30 [3.84]	0.26 [3.41]	0.24 [3.15]	0.23 [2.98]	0.22 [2.81]
Quintile	Cap	VW	0.21 [2.75]	0.22 [2.90]	0.20 [2.60]	0.17 [2.29]	0.19 [2.47]	0.17 [2.25]
Decile	NYSE	VW	0.29 [3.35]	0.28 [3.28]	0.24 [2.82]	0.22 [2.57]	0.24 [2.77]	0.22 [2.58]

Table 3: Conditional sort on size and SBS

This table presents results for conditional double sorts on size and SBS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SBS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SBS and short stocks with low SBS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	SBS Quintiles					SBS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.47 [1.91]	0.68 [2.65]	0.87 [3.40]	0.90 [3.85]	0.96 [4.11]	0.52 [7.23]	0.54 [7.54]	0.51 [7.14]	0.47 [6.48]	0.44 [6.11]	0.41 [5.65]
	(2)	0.62 [2.72]	0.68 [2.87]	0.85 [3.62]	0.96 [4.26]	0.91 [4.22]	0.30 [3.55]	0.34 [4.01]	0.28 [3.39]	0.26 [3.05]	0.24 [2.88]	0.23 [2.67]
	(3)	0.55 [2.74]	0.66 [3.05]	0.74 [3.39]	0.83 [4.09]	0.95 [4.84]	0.40 [5.34]	0.41 [5.40]	0.38 [5.01]	0.37 [4.70]	0.36 [4.58]	0.35 [4.38]
	(4)	0.51 [2.74]	0.65 [3.26]	0.76 [3.77]	0.77 [4.13]	0.79 [4.32]	0.28 [3.81]	0.29 [3.99]	0.23 [3.28]	0.22 [3.12]	0.12 [1.72]	0.13 [1.78]
	(5)	0.48 [2.93]	0.47 [2.62]	0.52 [2.99]	0.51 [3.09]	0.71 [4.52]	0.23 [2.55]	0.24 [2.59]	0.22 [2.33]	0.17 [1.85]	0.21 [2.24]	0.18 [1.88]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	SBS Quintiles					SBS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	383	384	384	382	374	31	34	40	29	29	
	(2)	109	107	108	108	107	57	56	58	57	55	
	(3)	78	78	78	77	78	99	96	98	100	100	
	(4)	66	65	66	66	66	206	204	215	213	217	
(5)	60	60	60	60	60	1479	1493	1661	1628	1887		

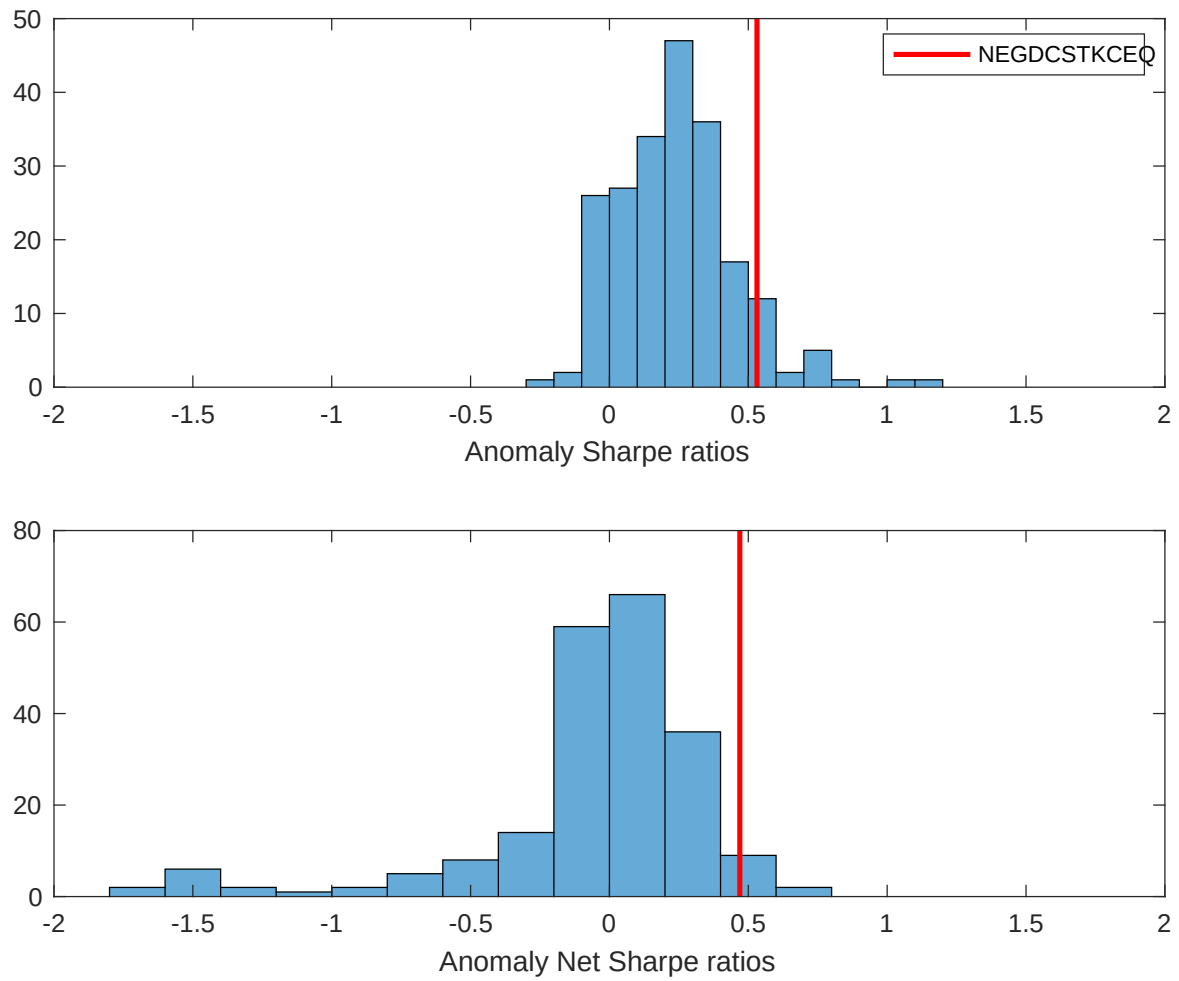


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SBS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

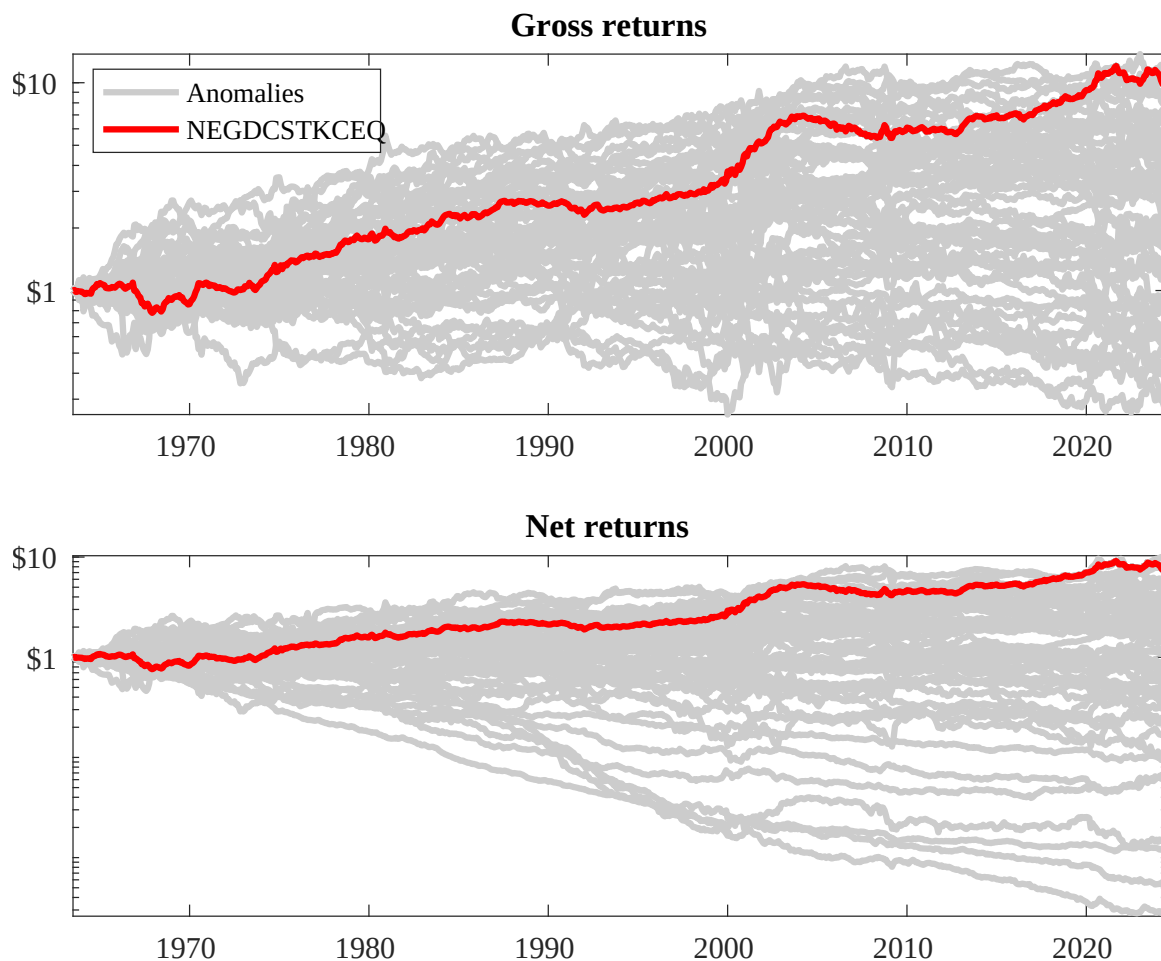
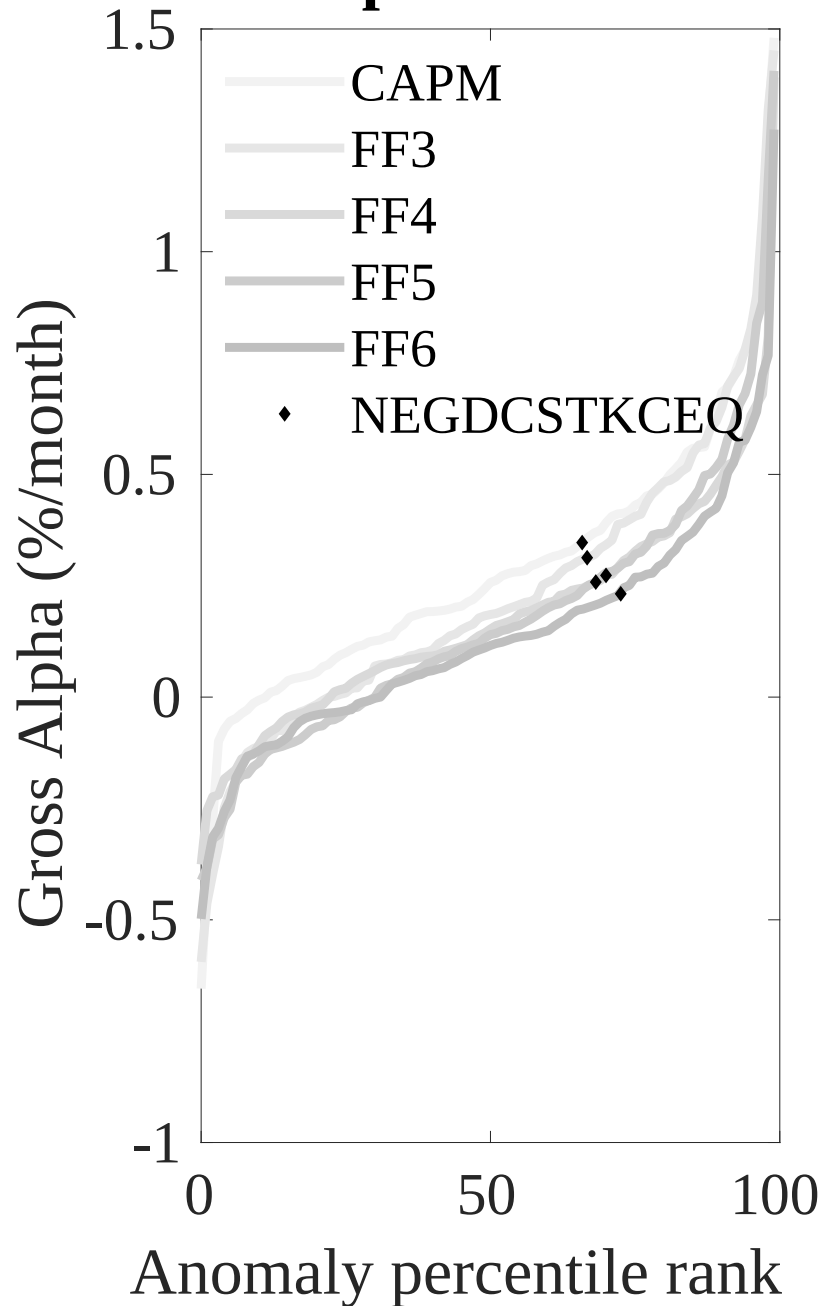


Figure 3: Dollar invested.

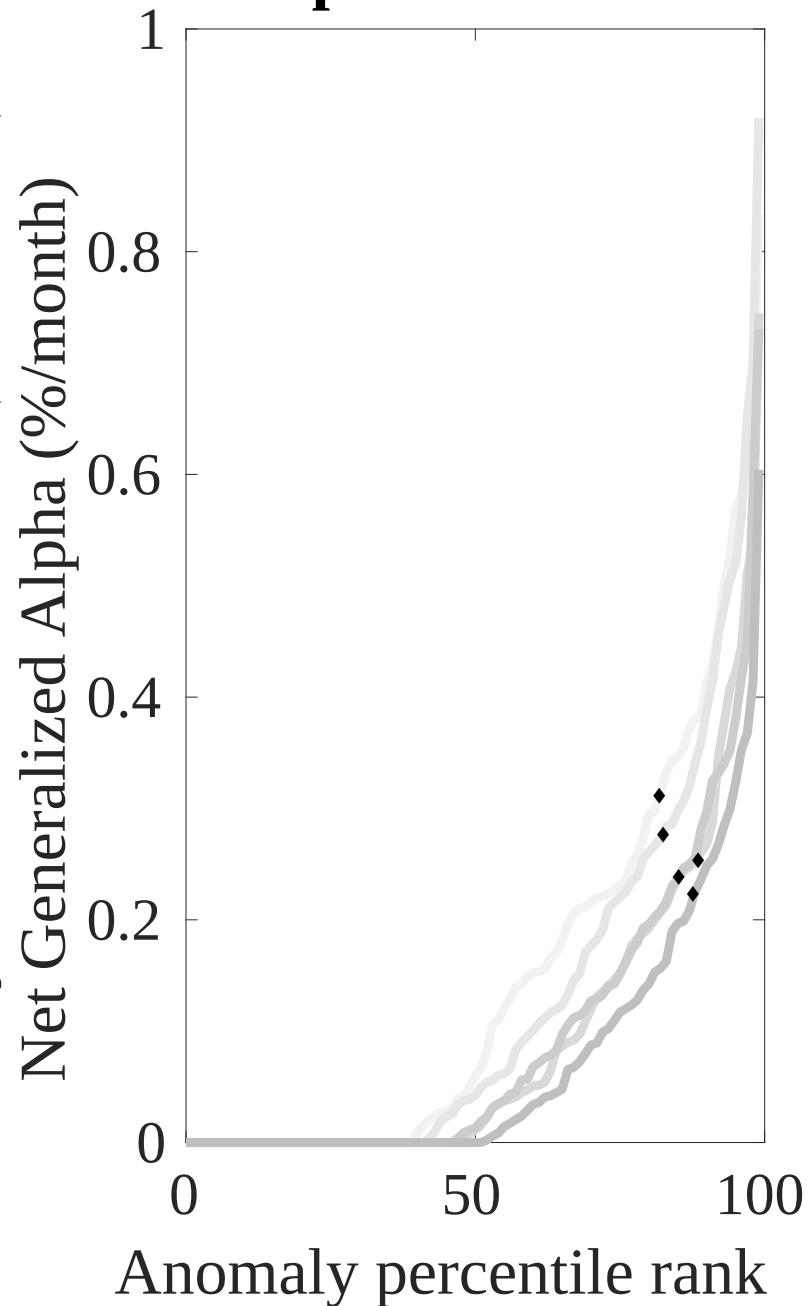
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SBS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



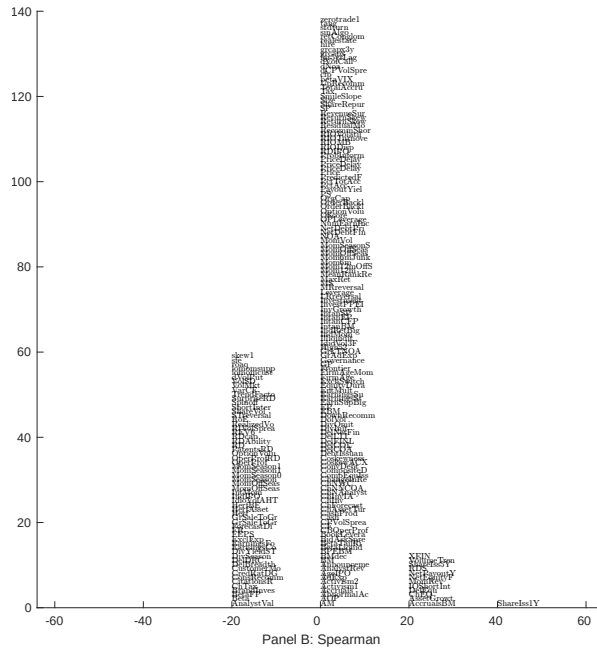
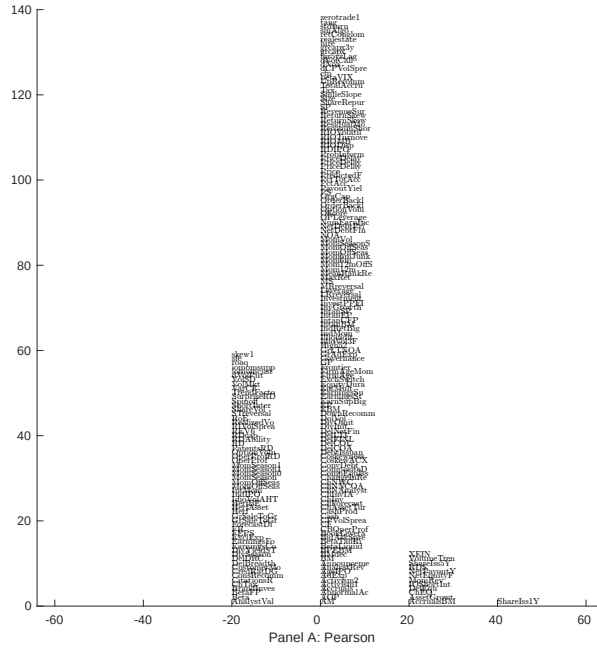


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with SBS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

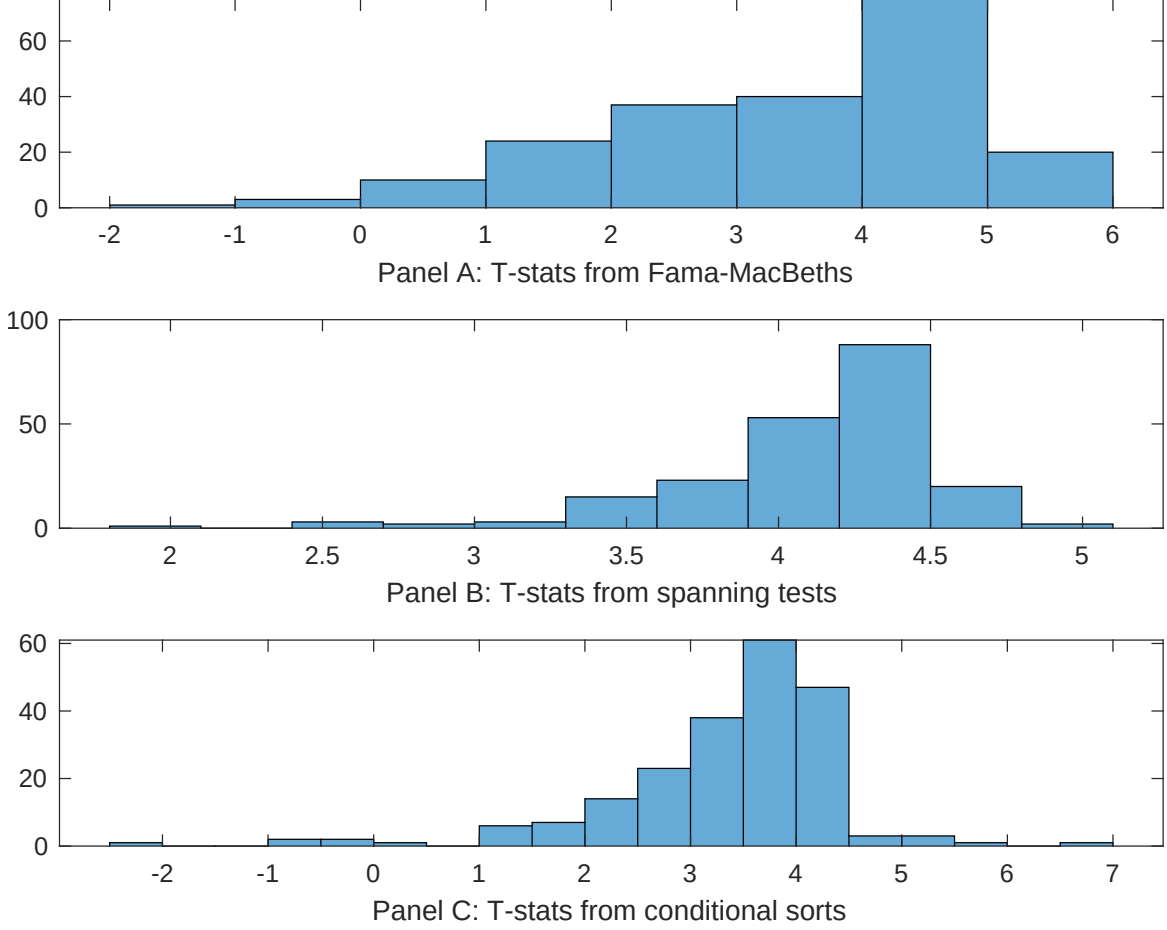


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of SBS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SBS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SBS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SBS. Stocks are finally grouped into five SBS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SBS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SBS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.04]	0.16 [6.83]	0.12 [5.43]	0.13 [6.43]	0.12 [5.68]	0.38 [1.12]	0.12 [2.73]
SBS	0.19 [3.45]	0.22 [3.64]	0.19 [2.68]	0.17 [3.07]	0.22 [3.72]	0.29 [0.85]	-0.42 [-1.43]
Anomaly 1	0.13 [5.75]						0.61 [1.21]
Anomaly 2		0.38 [3.18]					0.44 [1.83]
Anomaly 3			0.20 [1.80]				0.13 [1.48]
Anomaly 4				0.26 [4.71]			0.26 [2.32]
Anomaly 5					0.11 [3.08]		-0.13 [-1.56]
Anomaly 6						0.11 [4.27]	0.10 [3.48]
# months	715	720	715	715	720	648	510
$\bar{R}^2(\%)$	0	0	1	0	0	2	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SBS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SBS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.21 [2.79]	0.22 [2.86]	0.26 [3.32]	0.16 [2.02]	0.23 [2.96]	0.21 [2.65]	0.15 [2.00]
Anomaly 1	30.47 [7.88]						14.56 [3.14]
Anomaly 2		31.31 [7.56]					31.55 [5.39]
Anomaly 3			15.84 [5.24]				5.20 [1.60]
Anomaly 4				30.15 [7.35]			15.58 [3.38]
Anomaly 5					11.63 [2.97]		-17.59 [-3.28]
Anomaly 6						5.09 [4.77]	4.19 [4.14]
mkt	1.86 [1.02]	2.14 [1.17]	3.58 [1.87]	2.47 [1.34]	0.90 [0.48]	1.28 [0.68]	4.37 [2.41]
smb	4.54 [1.72]	2.25 [0.85]	5.88 [2.14]	5.35 [2.01]	3.15 [1.16]	1.32 [0.49]	3.60 [1.38]
hml	-2.40 [-0.68]	-5.40 [-1.52]	-7.69 [-2.02]	0.83 [0.23]	-3.07 [-0.84]	-2.81 [-0.77]	-5.82 [-1.61]
rmw	-18.17 [-4.40]	-0.17 [-0.05]	-10.87 [-2.69]	-12.47 [-3.23]	-0.58 [-0.16]	0.96 [0.26]	-16.23 [-3.76]
cma	9.97 [1.76]	-1.52 [-0.23]	15.49 [2.65]	14.26 [2.57]	17.10 [2.57]	26.36 [4.96]	-6.64 [-1.01]
umd	2.59 [1.43]	3.42 [1.89]	5.24 [2.82]	2.01 [1.10]	4.27 [2.29]	-0.97 [-0.46]	-2.39 [-1.18]
# months	728	732	728	728	732	728	728
$\bar{R}^2(\%)$	14	13	10	13	8	9	22

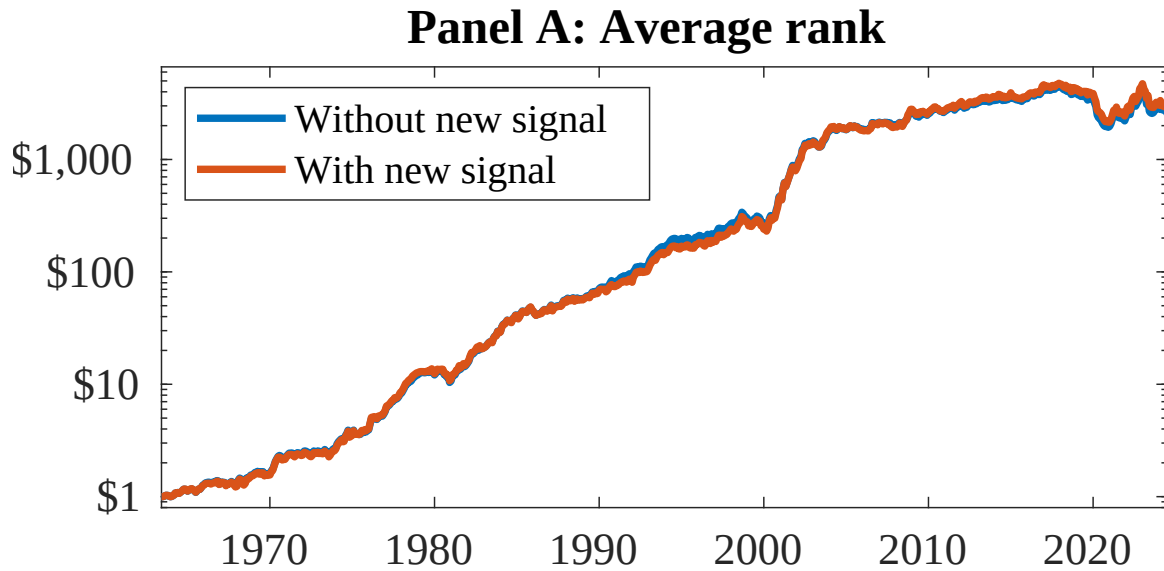


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as SBS. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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